

Set Abacus:

1. Solve

- a. We need to find the average CPI for **each segment** separately.

$$\text{Average CPI} = (\sum (\text{Instruction Count}_i * \text{CPI}_i)) / \text{Total Instruction Count}$$

$$\begin{aligned}\text{Average CPI}_{\text{Segment A}} &= (\sum (\text{Instruction Count}_i * \text{CPI}_i)) / \text{Total Instruction Count} \\ &= ((400,000 * 50\% * 2) + (400,000 * 30\% * 4) + (400,000 * 20\% * 3)) / 400,000 \\ &= (400,000 ((50\% * 2) + (30\% * 4) + (20\% * 3))) / 400,000 \\ &= (50\% * 2) + (30\% * 4) + (20\% * 3) \\ &= (0.5 * 2) + (0.3 * 4) + (0.2 * 3) \\ &= 1 + 1.2 + 0.6 \\ &= 2.8\end{aligned}$$

$$\begin{aligned}\text{Average CPI}_{\text{Segment B}} &= (\sum (\text{Instruction Count}_i * \text{CPI}_i)) / \text{Total Instruction Count} \\ &= ((600,000 * 40\% * 3) + (600,000 * 40\% * 2) + (600,000 * 20\% * 2)) / 600,000 \\ &= (600,000 ((40\% * 3) + (40\% * 2) + (20\% * 2))) / 600,000 \\ &= (40\% * 3) + (40\% * 2) + (20\% * 2) \\ &= (0.4 * 3) + (0.4 * 2) + (0.2 * 2) \\ &= 1.2 + 0.8 + 0.4 \\ &= 2.4\end{aligned}$$

- b. The program is divided into two segments. So the overall average CPI means CPI for the whole program(all segments).

$$\text{Average CPI} = (\sum (\text{Instruction Count}_i * \text{CPI}_i)) / \text{Total Instruction Count}$$

$$\text{Average CPI} = ((\text{Instruction Count}_{\text{Segment A}} * \text{CPI}_{\text{Segment A}}) + (\text{Instruction Count}_{\text{Segment B}} * \text{CPI}_{\text{Segment B}})) / \text{Total Instruction Count}$$

$$\begin{aligned}&= ((400,000 * 2.8) + (600,000 * 2.4)) / (400,000 + 600,000) \\ &= (1,120,000 + 1,440,000) / (1,000,000) \\ &= (2,560,000) / (1,000,000) \\ &= 2.56\end{aligned}$$

- c. Clock cycle = Instruction Count * CPI

$$\begin{aligned}\text{Total Clock cycle} &= \text{Total Instruction Count} * \text{Average CPI} \\ &= 1,000,000 * 2.56 \\ &= 2,560,000\end{aligned}$$

Alternate Solution:

$$\text{Total Clock cycle} = \sum (\text{Instruction Count} * \text{CPI})$$

$$\begin{aligned}
&= (400,000 * 50\% * 2) + (400,000 * 30\% * 4) + (400,000 * 20\% * 3) + (600,000 * 40\% * 3) + (600,000 * 40\% * 2) + (600,000 * 20\% * 2) \\
&= 2,560,000
\end{aligned}$$

d. Execution Time = Instruction Count * CPI * Clock Rate

$$\begin{aligned}
&= (1,000,000) * 2.56 / 1.536 * 10^9 \\
&= 0.00167s \\
&= 0.00167 * 10^3 \\
&= 1.67ms
\end{aligned}$$

e. Frequency_{old} = 1.536 GHz

After 20% increment, Frequency_{new} = 1.536 + (1.536 * 20%) GHz

$$= 1.8432GHz$$

As the CPI of memory instruction changes, the overall average CPI also changes.

CPI of memory instruction increased by 50%, so new CPI for memory instruction

$$= 4 + 4 * 50\% = 6$$

$$\text{Average CPI} = (\sum (\text{Instruction Count}_i * \text{CPI}_i)) / \text{Total Instruction Count}$$

$$\begin{aligned}
\text{Average CPI}_{\text{new}} &= ((\text{Instruction Count}_{\text{Segment A}} * \text{CPI}_{\text{Segment A}}) + (\text{Instruction Count}_{\text{Segment B}} * \text{CPI}_{\text{Segment B}})) / \text{Total Instruction Count} \\
&= ((400,000 * 50\% * 2) + (400,000 * 30\% * \mathbf{6}) + (400,000 * 20\% * 3) + (600,000 * 40\% * 3) + (600,000 * 40\% * 2) + (600,000 * 20\% * 2)) / 1,000,000 \\
&= (400,000 + 720,000 + 240,000 + 720,000 + 480,000 + 240,000) / 1,000,000 \\
&= 2.8
\end{aligned}$$

$$\begin{aligned}
\text{Execution Time}_{\text{new}} &= (\text{Instruction Count} * \text{CPI}) / \text{Clock Rate} \\
&= (1,000,000 * 2.8) / (1.8432 * 10^9) \\
&= 0.00152s \\
&= 0.00152 * 10^3 \\
&= 1.52 ms
\end{aligned}$$

The new execution time is 18.75% ((1.52 - 1.28) / 1.28 * 100%) more than the previous one.

f. MIPS = Clock Rate / (CPI * 10⁶)

$$\begin{aligned}
\text{MIPS}_{\text{previous}} &= 600 \\
\text{MIPS}_{\text{new}} &= 700
\end{aligned}$$

$$\begin{aligned}
\text{MIPS}_{\text{new}} &= 700 \\
\Rightarrow (\text{Clock Rate}_{\text{new}} / \text{CPI} * 10^6) &= 700 \\
\Rightarrow \text{Clock Rate}_{\text{new}} &= 700 * \text{CPI} * 10^6 \\
\Rightarrow \text{Clock Rate}_{\text{new}} &= 700 * 2.56 * 10^6
\end{aligned}$$

$$\Rightarrow \text{Clock Rate}_{\text{new}} = 1792000000$$

$$\text{Clock Rate}_{\text{new}} = 1.792 \text{ GHz}$$

$$g. \text{ MIPS} = \text{Clock Rate} / (\text{CPI} * 10^6)$$

$$\text{MIPS}_{\text{previous}} = 600$$

$$\text{MIPS}_{\text{new}} = 900$$

As the CPI of arithmetic instruction changes, the overall average CPI also changes.

CPI of arithmetic instruction increased by 50%, so new CPI for memory instruction = $2 + 2 * 20\% = 2.4$

$$\text{Average CPI} = (\sum (\text{Instruction Count}_i * \text{CPI}_i)) / \text{Total Instruction Count}$$

$$\begin{aligned} \text{Average CPI}_{\text{new}} &= ((\text{Instruction Count}_{\text{Segment A}} * \text{CPI}_{\text{Segment A}}) + (\text{Instruction Count}_{\text{Segment B}} * \text{CPI}_{\text{Segment B}})) / \text{Total Instruction Count} \\ &= ((400,000 * 50\% * \mathbf{2.4}) + (400,000 * 30\% * 4) + (400,000 * 20\% * 3) \\ &\quad + (600,000 * 40\% * 3) + (600,000 * 40\% * 2) + (600,000 * 20\% * 2)) / 1,000,000 \\ &= (480,000 + 480,000 + 240,000 + 720,000 + 480,000 + 240,000) / 1,000,000 \\ &= 2.64 \end{aligned}$$

$$\text{MIPS}_{\text{new}} = 900$$

$$\Rightarrow (\text{Clock Rate}_{\text{new}} / \text{CPI}_{\text{new}} * 10^6) = 900$$

$$\Rightarrow \text{Clock Rate}_{\text{new}} = 900 * \text{CPI}_{\text{new}} * 10^6$$

$$\Rightarrow \text{Clock Rate}_{\text{new}} = 900 * 2.64 * 10^6$$

$$\Rightarrow \text{Clock Rate}_{\text{new}} = 2376000000$$

$$\text{Clock Rate}_{\text{new}} = 2.376 \text{ GHz}$$

$$\text{Execution Time}_{\text{new}} = \text{Instruction Count} * \text{CPI}_{\text{new}} / \text{Clock Rate}_{\text{new}}$$

$$= (1,000,000 * 2.64) / (2.376 * 10^9)$$

$$= 0.001111111111 \text{ s}$$

$$= 1.11 \text{ ms}$$

$$\begin{aligned} \text{Percentage change in execution time} &= (1.67 - 1.11 / 1.28) * 100\% \\ &= 33.53\% \end{aligned}$$

$$h. \text{ Execution Time}_{\text{old}} = 1.67 \text{ ms (from d.)}$$

$$\text{Execution Time}_{\text{new}} = 1.67 - (1.67 * 10\%) = 1.67 - 0.167 = 1.503 \text{ ms}$$

Only the floating point part of the execution time benefits from the new additional cores, so only the execution time for floating point operations change. The rest of the execution time is not affected.

$$\begin{aligned} \text{Execution Time}_{\text{Floating Point}} &= (\text{IC} * \text{CPI}) / \text{CR} \\ &= (600,000 * 20\% * 2) / (1.536 * 10^9) \end{aligned}$$

$$= 0.15625\text{ms}$$

$$\text{Execution Time}_{\text{Rest Instructions}} = \text{Execution Time}_{\text{Arithmetic}} + \text{Execution Time}_{\text{Memory}} + \text{Execution Time}_{\text{Control}} + \text{Execution Time}_{\text{Logical}} + \text{Execution Time}_{\text{I/O}}$$

$$= (IC_{\text{Arithmetic}} * CPI_{\text{Arithmetic}}) / CR + (IC_{\text{Memory}} * CPI_{\text{Memory}}) / CR + (IC_{\text{Control}} * CPI_{\text{Control}}) / CR + (IC_{\text{Logical}} * CPI_{\text{Logical}}) / CR + (IC_{\text{I/O}} * CPI_{\text{I/O}}) / CR$$

$$\begin{aligned} &= (400,000 * 50\% * 2) / (1.536 * 10^9) + (400,000 * 30\% * 2) / (1.536 * 10^9) + (400,000 * 20\% * 3) / (1.536 * 10^9) + (600,000 * 40\% * 3) / (1.536 * 10^9) + (600,000 * 40\% * 2) / (1.536 * 10^9) \\ &= (400,000 + 240,000 + 240,000 + 720,000 + 480,000) / (1.536 * 10^9) \\ &= (2,080,000) / (1.536 * 10^9) \\ &= 1.3542\text{ms} \end{aligned}$$

As some part of the execution time is improved by adding additional cores, we can understand that this is an application of amdahl's law.

$$\begin{aligned} T_{\text{new}} &= T_{\text{affected}} / n + T_{\text{unaffected}} \\ \Rightarrow 1.503 &= 0.1562 / n + 1.3542 \\ \Rightarrow 0.1562 / n &= 0.1488 \\ n &= 1.05 \end{aligned}$$

So, 1.05 times speedup is achieved.

Set Byte

1. Solve

- a. We need to find the average CPI for **each segment** separately.

$$\text{Average CPI} = (\sum (\text{Instruction Count}_i * \text{CPI}_i)) / \text{Total Instruction Count}$$

$$\begin{aligned}\text{Average CPI}_{\text{Segment A}} &= (\sum (\text{Instruction Count}_i * \text{CPI}_i)) / \text{Total Instruction Count} \\ &= ((500,000 * 40\% * 2) + (500,000 * 35\% * 3) + (500,000 * 25\% * 3)) / 500,000 \\ &= (500,000 ((40\% * 2) + (35\% * 3) + (25\% * 3))) / 500,000 \\ &= (40\% * 2) + (35\% * 3) + (25\% * 3) \\ &= (0.4 * 2) + (0.35 * 3) + (0.25 * 3) \\ &= 0.8 + 1.05 + 0.75 \\ &= 2.60\end{aligned}$$

$$\begin{aligned}\text{Average CPI}_{\text{Segment B}} &= (\sum (\text{Instruction Count}_i * \text{CPI}_i)) / \text{Total Instruction Count} \\ &= ((500,000 * 30\% * 4) + (500,000 * 50\% * 4) + (500,000 * 20\% * 3)) / 600,000 \\ &= (500,000 ((30\% * 4) + (50\% * 4) + (20\% * 3))) / 500,000 \\ &= (30\% * 4) + (50\% * 4) + (20\% * 3) \\ &= (0.3 * 4) + (0.5 * 4) + (0.2 * 3) \\ &= 1.2 + 2.0 + 0.6 \\ &= 3.8\end{aligned}$$

- b. The program is divided into two segments. So the overall average CPI means CPI for the whole program(all segments).

$$\text{Average CPI} = (\sum (\text{Instruction Count}_i * \text{CPI}_i)) / \text{Total Instruction Count}$$

$$\text{Average CPI} = ((\text{Instruction Count}_{\text{Segment A}} * \text{CPI}_{\text{Segment A}}) + (\text{Instruction Count}_{\text{Segment B}} * \text{CPI}_{\text{Segment B}})) / \text{Total Instruction Count}$$

$$\begin{aligned}&= ((500,000 * 2.60) + (500,000 * 3.8)) / (500,000 + 500,000) \\ &= (1,300,000 + 1,900,000) / (1,000,000) \\ &= (3,200,000) / (1,000,000) \\ &= 3.20\end{aligned}$$

- c. Clock cycle = Instruction Count * CPI

$$\begin{aligned}\text{Total Clock cycle} &= \text{Total Instruction Count} * \text{Average CPI} \\ &= 1,000,000 * 3.20 \\ &= 3,200,000\end{aligned}$$

Alternate Solution:

$$\begin{aligned}
 \text{Total Clock cycle} &= \sum (\text{Instruction Count} * \text{CPI}) \\
 &= (500,000 * 40\% * 2) + (500,000 * 35\% * 3) + (500,000 * 25\% * 3) \\
 &\quad + (500,000 * 30\% * 4) + (500,000 * 50\% * 4) + (500,000 * 20\% * 3) \\
 &= 3,200,000
 \end{aligned}$$

d. $\text{Execution Time} = \text{Instruction Count} * \text{CPI} / \text{Clock Rate}$

$$\begin{aligned}
 &= (1,000,000 * 3.20) / (2 * 10^9) \\
 &= 0.0016\text{s} \\
 &= 0.0016 * 10^3 \\
 &= 1.6 \text{ ms}
 \end{aligned}$$

e. $\text{Frequency}_{\text{old}} = 2 \text{ GHz}$
 After 20% increment, $\text{Frequency}_{\text{new}} = 2 + (2 * 20\%) \text{ GHz}$
 $= 2.4 \text{ GHz}$

As the CPI of memory instruction changes, the overall average CPI also changes.

CPI of memory instruction increased by 50%, so new CPI for memory instruction
 $= 3 + 3 * 50\% = 4.5$

$$\text{Average CPI} = (\sum (\text{Instruction Count}_i * \text{CPI}_i)) / \text{Total Instruction Count}$$

$$\begin{aligned}
 \text{Average CPI}_{\text{new}} &= ((\text{Instruction Count}_{\text{Segment A}} * \text{CPI}_{\text{Segment A}}) + (\text{Instruction Count}_{\text{Segment B}} * \text{CPI}_{\text{Segment B}})) / \text{Total Instruction Count} \\
 &= ((500,000 * 40\% * 2) + (500,000 * 35\% * 4.5) + (500,000 * 25\% * 3) \\
 &\quad + (500,000 * 30\% * 4) + (500,000 * 50\% * 4) + (500,000 * 20\% * 3)) / 1,000,000 \\
 &= (400,000 + 787,500 + 375,000 + 600,000 + 1,000,000 + 300,000) / 1,000,000 \\
 &= 3.4625
 \end{aligned}$$

$$\begin{aligned}
 \text{Execution Time}_{\text{new}} &= (\text{Instruction Count} * \text{CPI}) / \text{Clock Rate} \\
 &= (1,000,000 * 3.4625) / (2.4 * 10^9) \\
 &= 0.00144 \text{ s} \\
 &= 0.00144 * 10^3 \\
 &= 1.44 \text{ ms}
 \end{aligned}$$

The new execution time is 10% $((1.6 - 1.44) / 1.44 * 100\%)$ less than the previous one.

f. $\text{MIPS} = \text{Clock Rate} / (\text{CPI} * 10^6)$
 $\text{MIPS}_{\text{previous}} = 625$
 $\text{MIPS}_{\text{new}} = 750$

$$\begin{aligned}
 \text{MIPS}_{\text{new}} &= 760 \\
 \Rightarrow (\text{Clock Rate}_{\text{new}} / \text{CPI} * 10^6) &= 750 \\
 \Rightarrow \text{Clock Rate}_{\text{new}} &= 750 * \text{CPI} * 10^6 \\
 \Rightarrow \text{Clock Rate}_{\text{new}} &= 750 * 3.20 * 10^6
 \end{aligned}$$

$$\Rightarrow \text{Clock Rate}_{\text{new}} = 2400000000$$

$$\text{Clock Rate}_{\text{new}} = 2.4 \text{ GHz}$$

$$g. \text{ MIPS} = \text{Clock Rate} / (\text{CPI} * 10^6)$$

$$\text{MIPS}_{\text{previous}} = 625$$

$$\text{MIPS}_{\text{new}} = 950$$

As the CPI of arithmetic instruction changes, the overall average CPI also changes.

CPI of arithmetic instruction increased by 50%, so new CPI for memory instruction = $2 + 2 * 20\% = 2.4$

$$\text{Average CPI} = (\sum (\text{Instruction Count}_i * \text{CPI}_i)) / \text{Total Instruction Count}$$

$$\begin{aligned} \text{Average CPI}_{\text{new}} &= ((\text{Instruction Count}_{\text{Segment A}} * \text{CPI}_{\text{Segment A}}) + (\text{Instruction Count}_{\text{Segment B}} * \text{CPI}_{\text{Segment B}})) / \text{Total Instruction Count} \\ &= ((500,000 * 40\% * \mathbf{2.4}) + (500,000 * 35\% * 3) + (500,000 * 25\% * 3) + (500,000 * 30\% * 4) + (500,000 * 50\% * 4) + (500,000 * 20\% * 3)) / 1,000,000 \\ &= (480,000 + 525,000 + 375,000 + 600,000 + 1,000,000 + 300,000) / 1,000,000 \\ &= 3.28 \end{aligned}$$

$$\text{MIPS}_{\text{new}} = 950$$

$$\Rightarrow (\text{Clock Rate}_{\text{new}} / \text{CPI}_{\text{new}} * 10^6) = 950$$

$$\Rightarrow \text{Clock Rate}_{\text{new}} = 950 * \text{CPI}_{\text{new}} * 10^6$$

$$\Rightarrow \text{Clock Rate}_{\text{new}} = 900 * 3.28 * 10^6$$

$$\Rightarrow \text{Clock Rate}_{\text{new}} = 2952000000$$

$$\text{Clock Rate}_{\text{new}} = 2.952 \text{ GHz}$$

$$\text{Execution Time}_{\text{new}} = \text{Instruction Count} * \text{CPI}_{\text{new}} / \text{Clock Rate}_{\text{new}}$$

$$= (1,000,000 * 3.28) / (2.952 * 10^9)$$

$$= 0.00111 \text{ s}$$

$$= 1.11 \text{ ms}$$

$$\begin{aligned} \text{Percentage change in execution time} &= (1.6 - 1.11 / 1.6) * 100\% \\ &= 30.63\% \end{aligned}$$

$$h. \text{ Execution Time}_{\text{old}} = 1.6 \text{ ms (from d.)}$$

$$\text{Execution Time}_{\text{new}} = 1.6 - (1.6 * 5\%) = 1.6 - 0.08 = 1.52 \text{ ms}$$

Only the floating point part of the execution time benefits from the new additional cores, so only the execution time for floating point operations change. The rest of the execution time is not affected.

$$\begin{aligned} \text{Execution Time}_{\text{Floating Point}} &= (\text{IC} * \text{CPI}) / \text{CR} \\ &= (500,000 * 20\% * 3) / (2 * 10^9) \end{aligned}$$

$$= 0.15 \text{ ms}$$

$$\text{Execution Time}_{\text{Rest Instructions}} = \text{Execution Time}_{\text{Arithmetic}} + \text{Execution Time}_{\text{Memory}} + \text{Execution Time}_{\text{Control}} + \text{Execution Time}_{\text{Logical}} + \text{Execution Time}_{\text{I/O}}$$

$$= (IC_{\text{Arithmetic}} * CPI_{\text{Arithmetic}}) / CR + (IC_{\text{Memory}} * CPI_{\text{Memory}}) / CR + (IC_{\text{Control}} * CPI_{\text{Control}}) / CR + (IC_{\text{Logical}} * CPI_{\text{Logical}}) / CR + (IC_{\text{I/O}} * CPI_{\text{I/O}}) / CR$$

$$\begin{aligned} &= (500,000 * 40\% * 2) / (2 * 10^9) + (500,000 * 35\% * 3) / (2 * 10^9) + (500,000 * 25\% * 3) / (2 * 10^9) + (500,000 * 30\% * 4) / (2 * 10^9) + (500,000 * 50\% * 4) / (2 * 10^9) \\ &= (400,000 + 525,000 + 375,000 + 600,000 + 1,000,000) / (2 * 10^9) \\ &= (2,900,000) / (2 * 10^9) \\ &= 1.45 \text{ ms} \end{aligned}$$

As some part of the execution time is improved by adding additional cores, we can understand that this is an application of amdahl's law.

$$\begin{aligned} T_{\text{new}} &= T_{\text{affected}} / n + T_{\text{unaffected}} \\ \Rightarrow 1.52 &= 0.15 / n + 1.45 \\ \Rightarrow 0.15 / n &= 0.07 \\ n &= 2.14 \end{aligned}$$

So, 2.14 times speedup is achieved.